ANANT GOEL

IoT / Robotics Software Intern

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SUMMARY

Final-year Computer Science student building software for connected devices and perception workflows. Project experience spans RFID inventory APIs, sensor telemetry, fog-edge simulation, vehicle tracking, and explainable road-risk logic.

EDUCATION & AVAILABILITY

BML Munjal University | B.Tech in Computer Science | Expected May 2027

Seeking on-site / hybrid internships in India. Flexible duration; start date to be discussed.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript

IoT software: Device REST APIs, RFID workflows, sensor logging, SQLite

Perception & simulation: OpenCV, YOLO, ByteTrack, calibrated geometry, virtual sensor nodes

Tools: Node.js, Express, Flask, Git, automated tests

SELECTED PROJECTS

RFID Smart Warehouse System | JavaScript, Node.js, Express, SQLite

- Built device-facing REST endpoints for RFID item registration, tag assignment, checkout, and temperature, humidity, gas, and alarm telemetry.
- Persisted inventory and sensor events in SQLite with timestamped audit logs and a web dashboard; designed the API for ESP32 devices using RC522, DHT22, and MQ sensors.

SpeedNet | Fog-Edge Simulation | Python, Flask, JSON telemetry

- Built deterministic virtual sensor nodes and a fog-layer API to calculate explainable highway speed advisories from weather and traffic observations.
- Validated sensor ranges, expired stale nodes, and exposed live status with thread-safe state; added reproducible scenarios and automated tests. Sensor inputs are simulated.

PathWise | Python, YOLO, ByteTrack, OpenCV

- Built a road-risk pipeline with persistent vehicle tracking, calibrated road-plane projection, and explainable time-to-collision and cut-in rules.
- Validated a six-second synthetic reference scenario containing 180 frames and 540 actor samples; demo outputs are not real-world perception-accuracy claims.